Milestone 2 – Feature Engineering & Baseline Modeling

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1. Objective

The objective of this milestone is to establish a high-performing baseline scanner source identification system using handcrafted feature engineering and classical machine learning models**.** This milestone focuses on extracting discriminative scanner-specific features and evaluating their effectiveness using baseline classifiers.

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2. Feature Engineering

Feature extraction is performed at the image level**,** where each document image is represented by a single feature vector.

2.1 Noise-Based Statistical Features

To emphasize scanner-specific artifacts, noise residuals are computed from grayscale images. From these residuals, the following statistical features are extracted:

- Mean
- Standard deviation
- Skewness
- Kurtosis
- Energy

2.2 Frequency-Domain Features (FFT)

Frequency-domain analysis is carried out by computing **radial Fast Fourier Transform (FFT) band energies** from the noise residual images.

- The Fourier spectrum is divided into multiple radial frequency bands
- The average energy within each band is calculated

2.3 Texture Features (Local Binary Patterns – LBP)

Local Binary Pattern (LBP) histograms are used to model fine-grained texture variations caused by different scanners.

- Uniform LBP encoding is applied
- Multiple radii are considered
- Normalized histograms are included as features

2.4 Feature Vector Construction

All extracted features—noise statistics, FFT band energies, and LBP texture features—are concatenated to form a single feature vector for each image. These feature vectors are stored in CSV format and used as input for baseline model training.

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3. Baseline Modeling

Two classical supervised machine learning models are used to evaluate baseline performance.

3.1 Random Forest Classifier

The Random Forest classifier is an ensemble-based model composed of multiple decision trees.

- Robust to feature correlations and noise
- Well-suited for heterogeneous handcrafted features

3.2 Support Vector Machine (SVM)

The Support Vector Machine classifier is implemented with a Radial Basis Function (RBF) kernel.

- Operates on normalized feature vectors
- Capable of learning non-linear decision boundaries

A **StandardScaler** is applied to normalize feature distributions prior to training and testing.

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4. Experimental Setup

- Dataset split: **80% training / 20% testing**

Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

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5. Results

5.1 Random Forest Results

The Random Forest classifier achieved an **overall accuracy of 98.03%** on the test set.

- Precision and recall values are consistently high across all scanner classes
- Most scanner classes achieve F1-scores between **0.98 and 0.99**
- HP and EpsonV550 scanners demonstrate near-perfect classification
- Minor confusion is observed among closely related Canon and Epson models

The macro-averaged and weighted F1-scores of **0.98** indicate balanced and reliable performance across all classes.

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•••  RF Classification Report

               precision    recall  f1-score   support

   Canon120-1       0.99      0.98      0.98        83
   Canon120-2       0.91      1.00      0.95        83
     Canon220       1.00      0.98      0.99        83
  Canon9000-1       1.00      0.98      0.99        83
  Canon9000-2       1.00      0.98      0.99        82
  EpsonV370-1       0.95      0.98      0.96        83
  EpsonV370-2       0.98      0.98      0.98        84
   EpsonV39-1       0.98      0.98      0.98        83
   EpsonV39-2       1.00      0.98      0.99        84
    EpsonV550       1.00      0.99      0.99        83
           HP       0.99      0.99      0.99        83

     accuracy                           0.98       914
    macro avg       0.98      0.98      0.98       914
 weighted avg       0.98      0.98      0.98       914

Overall Accuracy: 0.9803
```

5.2 Support Vector Machine Results

The SVM classifier achieved an **overall accuracy of 96.06%**.

- Precision and recall values mostly range between **0.91 and 1.00**
- Slightly higher confusion is observed among similar Canon scanner models
- EpsonV550 and HP continue to show strong classification performance

The macro and weighted F1-scores of **0.96** demonstrate good generalization capability.

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••• SVM Classification Report

                precision    recall  f1-score   support

    Canon120-1       0.98      0.95      0.96        83
    Canon120-2       0.84      1.00      0.91        83
      Canon220       1.00      0.95      0.98        83
   Canon9000-1       1.00      0.95      0.98        83
   Canon9000-2       1.00      0.95      0.97        82
   EpsonV370-1       0.91      0.95      0.93        83
   EpsonV370-2       1.00      0.95      0.98        84
    EpsonV39-1       1.00      0.95      0.98        83
    EpsonV39-2       0.91      0.95      0.93        84
     EpsonV550       1.00      0.98      0.99        83
            HP       0.98      0.98      0.98        83

      accuracy                           0.96       914
     macro avg       0.96      0.96      0.96       914
  weighted avg       0.96      0.96      0.96       914

Overall Accuracy: 0.9606
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5.3 Comparative Performance

Model	Accuracy
Random Forest	**98.03%**
Support Vector Machine	**96.06%**

6.Testing

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Image: /content/drive/MyDrive/Tracefinder/processed/images/Wikipedia_Canon9000-1_300_s4_30.png
True Label : Canon9000-1
RF Predict : Canon9000-1
SVM Predict: Canon9000-1
--------------------------------------------------
```

True: Canon9000-1 | RF: Canon9000-1 | SVM: Canon9000-1

```
Image: /content/drive/MyDrive/Tracefinder/processed/images/Wikipedia_Canon120-1_150_s1_55.png
True Label : Canon120-1
RF Predict : Canon120-1
SVM Predict: Canon120-1
--------------------------------------------------
```

True: Canon120-1 | RF: Canon120-1 | SVM: Canon120-1

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6. Conclusion

This milestone successfully establishes a high-accuracy baseline scanner identification system using handcrafted feature engineering and classical machine learning models. The achieved accuracies of **98.03% (Random Forest)** and **96.06% (SVM)** validate the effectiveness of the extracted features and provide a strong foundation for further research and enhancement in subsequent milestones.
